

SHRINIVAS A MASALAVAD

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PROFILE SUMMARY

AI, Embedded Systems, and Computer Vision engineer with hands-on experience building **8+ AI and embedded engineering projects** that span precision agriculture, IoT, wireless networking, automation, and intelligent surveillance systems.

Currently a Research Intern at **IIT Dharwad**, working on embedded systems, wireless communication, and distributed edge computing, with prior training as an AI & Embedded Systems Trainee at **NITK Surathkal**, building practical experience in embedded systems, Raspberry Pi, wireless communication, and edge computing.

Presented technology-driven solutions at **8+ national, state, and district-level innovation exhibitions**, including a New Delhi (2024) showcase to researchers, academicians, and government representatives; work recognized by the **Department of Science and Technology (DST), Government of India** for its real-world impact.

Skilled in Python, C/C++, OpenCV, YOLO, Raspberry Pi, ESP32, LoRa, IoT, and Edge AI, delivering end-to-end solutions for smart agriculture, wildlife monitoring, and automation — with a strong focus on solving scalable, real-world engineering challenges across AI/ML, Computer Vision, Embedded Systems, IoT, and Wireless Networking.

EXPERIENCE

Research Intern | *Indian Institute of Technology Dharwad (IIT Dharwad)* | Apr 2026 – Present | Dharwad, Karnataka

- Contributing to an active research project focused on embedded systems, wireless communication, and distributed edge computing.
- Designed and deployed a secure image transmission system between **2 Raspberry Pi devices** operating across different networks, enabling reliable data exchange across distributed edge nodes.
- Developed a Python-based image transfer pipeline enabling reliable, secure, and low-latency communication between distributed edge devices.
- Configured remote connectivity protocols — including SSH-based access and network tuning — and optimized network communication for cross-network image transmission.
- Performed **50+ system validation and debugging sessions**, working closely with faculty researchers to systematically identify issues and improve overall communication reliability.

AI & Embedded Systems Trainee | *National Institute of Technology Karnataka (NITK), Surathkal* | May 2023 | Mangalore, Karnataka

- Selected for hands-on research training on an AI-powered Intelligent Weed Recognition and Laser-Based Eradication System for precision agriculture, working directly under faculty/professor guidance.
- Studied the system's research background and contributed to its architecture and technical design, developing a working understanding of how AI, embedded hardware, and mechanical components come together in a single system.
- Gained exposure to the full development lifecycle of the project — including simulation, mechanical/SolidWorks design, hardware integration, and software development workflows — as part of the training program.
- Built foundational experience with ROS 2 and Gazebo for robotic simulation, system architecture, and virtual testing, alongside SolidWorks for mechanical design.
- Developed a practical understanding of AI model development and computer vision workflows, and how they integrate with embedded system architecture, through direct involvement in the project's development process.

Technologies: Python • OpenCV • ESP32 • ROS 2 • Gazebo • SolidWorks • Computer Vision • Embedded Systems

PROJECTS

Intelligent Weed Recognition and Laser-Based Eradication System

- Built an AI-powered weed recognition and laser-based eradication system for precision agriculture, trained and evaluated on **1000+ annotated agricultural images** to reliably distinguish weeds from crops.
- Designed an ESP32-based embedded control system integrating camera and actuator hardware for real-time weed detection and automated response.

- Implemented an end-to-end computer vision pipeline in Python and OpenCV, validated through **50+ prototype experiments** to confirm detection accuracy and system reliability under field-like conditions.
- Delivered a functional precision-agriculture solution combining AI-based weed detection with automated laser-based eradication, designed for real-world field use.

Smart Laser-Based Elephant Intrusion Detection System — *KSSC / DST, Government of Karnataka*

- Designed a laser-based virtual perimeter system for early elephant intrusion detection in coffee plantations and farmland, aimed at reducing human-wildlife conflict.
- Built an embedded monitoring system with ESP32, Laser/IR sensors, and **3 communication modules (LoRa, GSM, Wi-Fi)** to support reliable real-time alerts even in areas with limited connectivity.
- Engineered an alert pipeline notifying farmers instantly via SMS and a mobile app across **2 channels**, ensuring reliable remote monitoring and faster response times.
- Selected through KSSC evaluation to present at the Karnataka State-Level Science Exhibition, engaging international delegates from Germany and Nigeria.

Smart Solar-Powered Bird Repellent System for Crop Protection — *Embedded Systems / Renewable Energy*

- Built a solar-powered autonomous bird repellent system with a **6000 mAh battery** for continuous, self-sustained operation even in low-sunlight conditions.
- Implemented time-based automation (5:00 AM–6:00 PM) so the system only runs during active deterrence hours, cutting unnecessary power consumption by **~11 hours/day** and extending battery life.
- Programmed rotating deterrent-sound sequences every 30 minutes to reduce bird habituation to a single repeated sound and improve long-term effectiveness.

Pick & Place Robotic Arm — *2024*

- Designed a 4-joint robotic arm in Fusion 360 and 3D-printed the components to sub-0.5mm tolerance for precise, repeatable assembly.
- Programmed inverse-kinematics (IK) based servo control for repeatable end-effector positioning across a **180° workspace**, enabling accurate pick-and-place motion.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL

AI & Computer Vision: Machine Learning, Deep Learning, Computer Vision, OpenCV, YOLO, Image Processing, Edge AI

Embedded Systems & IoT: ESP32, Raspberry Pi, Arduino, Embedded C, IoT, Sensor Integration, GPIO, UART, I2C, SPI

Robotics & Simulation: ROS 2, Gazebo, SolidWorks, Robotics Fundamentals

Networking & Edge Computing: TCP/IP, SSH, Remote Connectivity, Wireless Communication, LoRa, Edge Computing

Tools & Frameworks: Git, GitHub, Linux, VS Code, Jupyter Notebook

Cloud & Databases: Firebase, MySQL (Basic)

EDUCATION

Master of Computer Applications (MCA) | Expected 2028

Bengaluru University, Bengaluru | Pursuing

Bachelor of Computer Applications (BCA) | Graduated 2026

Karnataka University, Dharwad | CGPA: 8.3/10

CERTIFICATIONS & ACHIEVEMENTS

- **Machine Learning Specialization** — Coursera (DeepLearning.AI & Stanford University), covering Supervised Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommender Systems, and Reinforcement Learning.
- Selected for **2 National-Level, 2 State-Level, and 4 District-Level Science & Innovation Exhibitions (8+ total)**, presenting AI, embedded systems, and precision-agriculture research to technical and government audiences.
- Presented projects alongside international delegates from Germany and Nigeria, fostering technical knowledge exchange with global researchers and industry representatives.
- Innovation project appreciated by the **Department of Science and Technology (DST), Government of India** for its real-world impact on precision agriculture and wildlife-conflict mitigation.